

Emily Herbert

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Education

- 2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.
- 2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.
- 2025 **Art & Design Intensive**, *Atelier de Sevres*, Paris, France.

Work Experience

- Aug 2021 - **Fuel Labs**.
- May 2023 Compiler Engineer - Rust
Designed and implemented a compiler for the Sway smart contract programming language, focusing on friendly interactivity with developers. Built tools to integrate with the FuelVM and the Fuel ecosystem.
- May 2021 – **Google**, *Madison, WI*.
- Aug 2021 Software Engineering Intern
Implemented load balancing in a library meant to interface with the network card and perform RPC-like operations using RMA, achieved by integrating two existing early-development libraries together. Contributed to app design on a Google 2023 project.
- June 2017 – **National Aeronautics and Space Administration (NASA)**, *Langley, VA*.
- Aug 2017 NASA Internships, Fellowships, and Scholarships (NIFS) Intern
Contributed to the NASA Safeguard autonomous drone geofencing project. Designed and implemented system for on-board flight control of GPS devices. Refactored code from previous NASA flight missions to meet current mission standards.
- June 2016 – **General Electric (GE), Oil & Gas**, *Billerica, MA*.
- Aug 2016 Information Technology Leadership Program (ITLP) Intern
Created asset tracking system for shop floor using RFID, Bluetooth LE, and Raspberry Pi. Worked with the SAP enterprise resource management software to automate EHSM compliance checks.

Research Experience

- Jan 2021 - **NeuPRL Lab**, *Northeastern University*.
- May 2021 Northeastern Programming Languages Lab (NeuPRL), advised by Prof. Arjun Guha. Researching programming language and systems tools for serverless computing.
prl.ccs.neu.edu
- May 2019 - **PLASMA Lab**, *University of Massachusetts Amherst*.
- Jan 2021 Programming Languages and Systems at Massachusetts lab (PLASMA), advised by Prof. Arjun Guha. Researching programming language and systems tools for serverless computing.
plasma-umass.org
- June 2018 - **DREAM Lab**, *University of Massachusetts Amherst*.
- May 2019 Data systems Research for Exploration, Analytics, and Modeling lab (DREAM), advised by Prof. Peter Haas. Researching deep learning methods for simulation input modeling.
dbgroup.cs.umass.edu

Teaching Experience

- Sep 2018 - **University of Massachusetts Amherst**, *Amherst, MA*.
- May 2019 Programming Methodology *Teaching Assistant*, Mathematical Foundation for Informatics *Teaching Assistant*
- Aug 2016 - **Trinity University**, *San Antonio, TX*.
- May 2018 Principles of Computer Science II *Teaching Assistant*, Introduction to Programming Logic *Teaching Assistant*, Principles of Computer Science II *Teaching Assistant*

Awards & Scholarships

- July 2019 **David W. Stemple Scholarship in Computing**, *University of Massachusetts Amherst*.
Provides support to a first-year graduate student in Computer Science pursuing a Ph.D. in Systems research.
- May 2019 **UMass CICS Women's Travel Grant**, *University of Massachusetts Amherst*.
Awarded to UMass CICS women to assist with conference travel expenses.
- May 2019 **UMass CICS Travel Grant**, *University of Massachusetts Amherst*.
Awarded to UMass CICS students to assist with conference travel expenses.
- April 2019 **PLISS Studentship**.
Awarded to new programming languages and systems researchers to attend the 2019 Programming Languages Implementation Summer School (PLISS).
- Nov 2018 **PLMW and POPL 2019 Scholarship**, *SIGPLAN*.
Awarded to new programming languages researchers to attend the 2019 Programming Languages Mentoring Workshop (PLMW) and Principles of Programming Languages Conference (POPL).
- Mar 2018 **2018 UMass CICS Fellowship**, *University of Massachusetts Amherst*.
Awarded to an outstanding graduate student applicant within the UMass CICS applicant pool.

Publications

- Wang Cen, Emily A Herbert, and Peter J Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference*. 2020. [[paper](#)]
Best Contributed Theoretical Paper Finalist
- Emily A Herbert, Wang Cen, and Peter J Haas. NIM: Generative Neural Networks for Simulation Input Modeling. *Summer Simulation Conference*. 2019. [[short paper](#)]

Preprints

- Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. 2021. [[preprint](#), [repo](#)]

Talks & Panels

- Blockchain Language Design panel. *Layer 2 Day at EthDenver*. 2023. [[recording](#)]
- Introduction to Fuel: Let's Get Modular talk. *Celestia's Modular Fellows*. 2022. [[slides](#)]
- Beyond Monolithic with Fuel: the Fastest Modular Execution Layer talk. *HackMoney*. 2022. [[slides](#)]
- Scaling Execution: Optimistic panel. *The Modular Summit*. 2022.
- Developing Smart Contracts in Sway talk talk. *Layer2Amsterdam*. 2022. [[slides](#)]
- Sway: A Rust-based Smart Contract Language talk. *EthDenver*. 2022. [[slides](#)]
- The Rollup Developer Experience panel. *EthDenver*. 2022. [[recording](#)]
- My Time From Trinity Until Now. *Trinity University*. 2021.
- A Language-based Serverless Function Accelerator talk. *Google PhD Intern Research Conference*. 2021.
- Graduate Student Life. *Trinity University*. 2020.
- A Language-based Serverless Function Accelerator talk. *Cornell CAPRA Lab*. 2020. [[slides](#)]
- NIM: Generative Neural Networks for Simulation Input Modeling talk. *SCS Summer Simulation Conference*. 2019. [[slides](#)]

Service

- 2021 EuroSys 2021 Shadow PC Member

- 2020 Mentor to 3 first-year UMass CICS PhD students
UMass CICS PhD applicant support program reviewer
Northeastern University PhD applicant support program reviewer
#ShutdownPL @ ICFP volunteer
ShutdownPL volunteer
- 2019 Mentor to 1 first-year UMass CICS PhD student
Winter Simulation Conference poster session reviewer